

# EE8257 Information Security



## Lecture 3

### Introduction to cryptography and basic cryptographic methods

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# Outline

- Introduction to Cryptography
- Crypto Methods Overview
- Simple Substitution – Shift base ciphers
- Simple Substitution – Permutation base ciphers
- Cryptanalysis Overview
- Double Transposition Cipher
- One - Time Pad
- Confusion and Diffusion
- Exhaustive key search

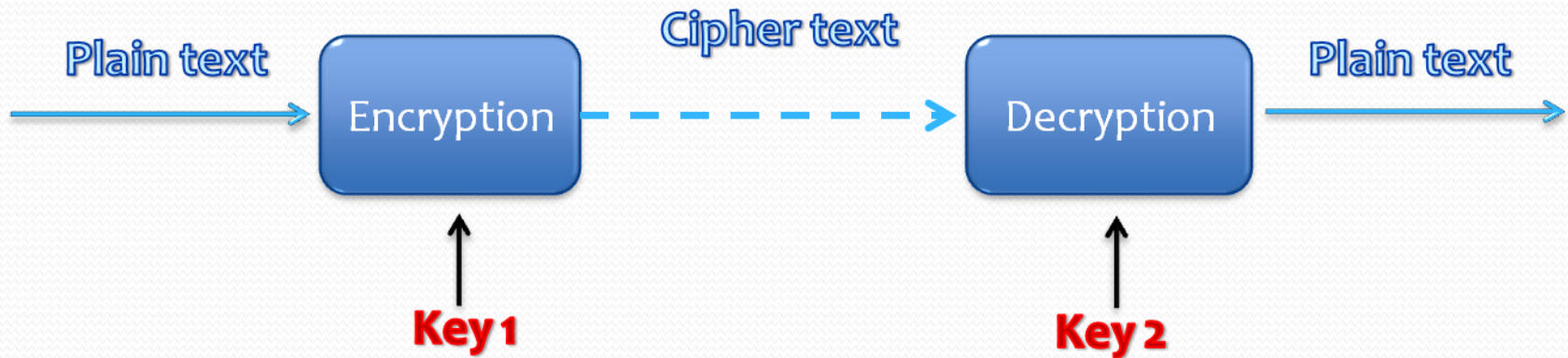


# Introduction to Cryptography

- Cryptography – Practice and study of techniques for secure communication
- Cryptanalysis – The study of breaking and revealing secret codes produced through cryptography
- Encrypting plaintext to produce cipher text – secret code
- **Encryption** is the process of encoding messages or information in such a way that only authorized parties can read it [Wikipedia].
- Assuring Confidentiality, Integrity, Availability, Authenticity and Privacy....



# Introduction to Cryptography (Cont...)



- A cipher or cryptosystem is used to encrypt the plaintext
- The result of encryption is ciphertext
- We decrypt ciphertext to recover plaintext
- A key is used to configure a cryptosystem
- A symmetric key cryptosystem uses the same key to encrypt as to decrypt (Key 1 = Key 2)
- A **public key** cryptosystem uses a public key to encrypt and private key to decrypt (Key 1  $\neq$  Key 2)

# Introduction to Cryptography (Cont...)

- Basis assumption of cryptography:
  - The crypto system is completely known to the public
  - Only the **key(s)** is/are secret
- Also known as *Kerckhoffs Principle*
  - Crypto algorithms are not secret
- Why do we make this assumption?
  - Experience has shown that secret algorithms are weak when exposed
  - Secret algorithms never remain secret
  - Better to find weaknesses beforehand
  - More efficient and secure systems could be presented when analyzing for all possible attacks and vulnerabilities

# Crypto Methods Overview

- Simple Substitution
- Symmetric Key Encryption
- Asymmetric Key Encryption
- Hash Algorithms

# Simple Substitution

- Encrypting messages by substituting letters, numbers and symbols to elements of the original message
  - E.g.: message = “ I am Alice ”  
  
“ ” = 000, A = 001, a = 101, B = 002, b = 102, C = 003, c = 103...  
.....e = 105, I = 009, i = 109, l = 112, m = 113
  - Cipher text → 009000101113000001009109003105
  - Cipher text could be decrypted by reversing the process
  - Keys are the letters and their corresponding substitutions

# Simple Substitution (Cont...)

- Shift base Cipher : Substitution by shifting the alphabet
  - Key is the shifting number
- Shift by 4 :

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
| W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |

- Encrypt the text : I am Alice → E WI WHEYA
- Try decrypting the cipher text : “E WI WJ EZEKP”
- Shift by 3 is called a ***Caesar’s cipher***
  - Decrypt “FDHVDUV FLSKHU”



# Simple Substitution (Cont...)

- Non simple substitution
  - Key  $n = \{0,1,2,\dots,25\}$
- If the key is not given → try all the possibilities until you determine the key (26 possibilities) → **Exhaustive Key Search**
- Try deciphering the cipher text : “ZGEGEGJHZAKE”
- Developing a substitution system :
  - Decrypting for all possible keys
  - Match the decrypted words with the known dictionaries to determine the key

# Simple Substitution (Cont...)

- Even less simple substitution
- Key is a permutation of letters  $\rightarrow {}^nP_r$
- If  ${}^{26}P_r \rightarrow r$  could be 26
  - Therefore, possibilities  $\rightarrow 26! = 2^{88}$
- Ciphertext :

PBFPVYFBQXZTYFPBFEQJHDXXQVAPTPQJKTOYQWIPBVWLXTOXBTFXQWAXBVCXQW  
AXFQJ VWLEQNTQZQGGQLFXQWAKVWLXQWAEBIPBFXFQVXGTVJVWLBTPQWAEBFP  
BFHCVLXBQUFEVWLXGDPEQVPQGVPPBFTIXPFHXZHVFAGFOTHFEFBQUFTD HZBQP  
OTHXTYFTODXQHFTDPTOGHFQPBQWAQJJTODXQHFOQPWTBDHHIXQVAPBFZQHC  
FWPFHPBFIPBQWKFABVYYDZBOTHBPBPQJQTQOTOGHFQAPBFEQJHDXXQVAVXEBQP  
EFZBVFOJIWFFACCFCHQWAUVWFLQHGFXXVAFXQHUFUHFILTTAVWAFFAWTEVOITD  
HFHFQAITIXPFHXAFQHEFZQWGFLVWPTOFFA

# Simple Substitution (Cont...)

PBFPVYFBQXZTYFPBFEQJHDXXQVAPTPOJKTOYQWIPBVWLXTOXBTFXQWAXBVCXQWAXFOJWVLEQNTQZQGGQLFXQWAKVWLX  
QWAEIBIPBFXFQVXGTVJVWLBTPQWAEIBFPBFHCVLXBQUFVWLXGDPEQVPOQVPPBFTIXPFHXZHVFAGFOTHFEFBQUFTDHBZBQPOT  
HXTYFTODXQHFTDPTOGHFQBPBQWAQJJTODXQHFOQPWTBDHHIXQVAPBFZQHCFWPFHPBFIPBQWKFABVYYDZBOTHBPBQJTO  
OTOGHFQAPBFEQJHDXXQVAVXEBQPEFZBVFOJIWFFACFCFHQWAVVWFLQHGXVAFXQHUFUHLTTAVWAFFAWTEVOITDHFHF  
QAITIXPFHXAQHEFZQWGFLVWPTOFFA

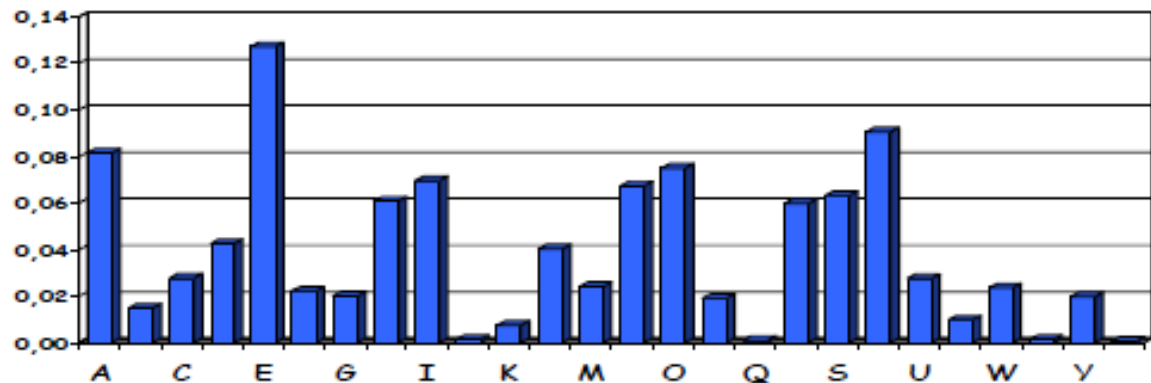
- English letter frequency count
- Compare the ciphertext element frequencies with the common usage of letter frequencies of English language
  - Letters such as 'e', 't' and 'a' are more commonly used than others

OR consider the letter frequency set as a pattern and try to identify the pattern in common sets with similar word count → Then the sample space would become larger

| A  | B  | C | D  | E  | F  | G  | H  | I  | J | K | L  | M | N | O  | P  | Q  | R | S | T  | U | V  | W  | X  | Y | Z |
|----|----|---|----|----|----|----|----|----|---|---|----|---|---|----|----|----|---|---|----|---|----|----|----|---|---|
| 21 | 26 | 6 | 10 | 12 | 51 | 10 | 25 | 10 | 9 | 3 | 10 | 0 | 1 | 15 | 28 | 42 | 0 | 0 | 27 | 4 | 24 | 22 | 28 | 6 | 8 |

Ciphertext element count

English letter count



# Cryptanalysis Overview

- Cryptosystem is **secure** if best known attack is to try all the keys → **Exhaustive key search**
- Cryptosystem is **insecure** if any **shortcut attack** is known
  - The algorithm becomes vulnerable
- By this definition, an **insecure system** might be harder to break than a secure system!
  - secure cryptosystem – attacker need to determine one or two keys
  - insecure cryptosystem with more keys – attacker has to reveal all the keys → more work
  - Having more keys would make the cryptosystem inefficient, even though it is more secure
- Best approach :
  - **Optimized number of keys (2 or 3)**
  - Keys with large enough **key space** (128-bit →  $2^{128}$ )

# Double Transposition Cipher

- Write the plaintext into a matrix
- Permute the rows and columns of the matrix
- Derive the ciphertext from the matrix
- Plain text : I am an Engineer → IxamxanxEngineer

|   | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | I | x | a | m |
| 2 | x | a | n | x |
| 3 | E | n | g | i |
| 4 | n | e | e | r |



|   | 4 | 3 | 1 | 2 |
|---|---|---|---|---|
| 1 | m | a | I | x |
| 3 | i | g | E | n |
| 2 | x | n | x | a |
| 4 | r | e | n | e |

- Key : matrix size and permutation ( $[1324]$ ,  $[4312]$ )
- Cipher Text : maixigenxnxarene
- Try decrypting → TGNxHBGIxAYOERxxHTEB ; ( $[3241]$ ,  $[42135]$ )

# One Time Pad (OTP)

- Also known as *Vernam cipher*
- Cipher is based on the ***XOR operation*** of the bits
- In other words → Modulo 2 addition (mod 2)

| A | B | XOR |
|---|---|-----|
| 0 | 0 | 0   |
| 0 | 1 | 1   |
| 1 | 0 | 1   |
| 1 | 1 | 0   |

- Plaintext is represented by a bit-stream
- Key → a bit-stream of length equal to plaintext bit-stream

• Encryption →  $\text{Plaintext}_{\text{bit-stream}} \oplus \text{Key}_{\text{bit-stream}}$

• Decryption →  $\text{Ciphertext}_{\text{bit-stream}} \oplus \text{Key}_{\text{bit-stream}}$

# One Time Pad (OTP) (Cont...)

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Encryption :  $\text{Plaintext}_{\text{bit-stream}} \oplus \text{Key}_{\text{bit-stream}}$

|             |     |     |     |     |     |     |     |     |     |     |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|             | h   | e   | i   | l   | h   | i   | t   | l   | e   | r   |
| Plaintext:  | 001 | 000 | 010 | 100 | 001 | 010 | 111 | 100 | 000 | 101 |
| Key:        | 111 | 101 | 110 | 101 | 111 | 100 | 000 | 101 | 110 | 000 |
| Ciphertext: | 110 | 101 | 100 | 001 | 110 | 110 | 111 | 001 | 110 | 101 |
|             | s   | r   | l   | h   | s   | s   | t   | h   | s   | r   |

# One Time Pad (OTP) (Cont...)

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

Decryption : Ciphertext<sub>bit-stream</sub>  Key<sub>bit-stream</sub>

|             | s   | r   | l   | h   | s   | s   | t   | h   | s   | r   |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ciphertext: | 110 | 101 | 100 | 001 | 110 | 110 | 111 | 001 | 110 | 101 |
| Key:        | 111 | 101 | 110 | 101 | 111 | 100 | 000 | 101 | 110 | 000 |
| Plaintext:  | 001 | 000 | 010 | 100 | 001 | 010 | 111 | 100 | 000 | 101 |
|             | h   | e   | i   | l   | h   | i   | t   | l   | e   | r   |



# One Time Pad (OTP) (Cont...)

- If an improper / false ‘key’ is used (case 1)
  - Key 1 = 1011110001011111000000101110000

|              | s   | r   | l   | h   | s   | s   | t   | h   | s   | r   |
|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ciphertext:  | 110 | 101 | 100 | 001 | 110 | 110 | 111 | 001 | 110 | 101 |
| “key”:       | 101 | 111 | 000 | 101 | 111 | 100 | 000 | 101 | 110 | 000 |
| “Plaintext”: | 011 | 010 | 100 | 100 | 001 | 010 | 111 | 100 | 000 | 101 |
|              | k   | i   | l   | l   | h   | i   | t   | l   | e   | r   |

# One Time Pad (OTP) (Cont...)

- If an improper / false 'key' is used (case 2)
  - Key 2 = 111101000011101110001011101101

|              | s   | r   | l   | h   | s   | s   | t   | h   | s   | r   |
|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Ciphertext:  | 110 | 101 | 100 | 001 | 110 | 110 | 111 | 001 | 110 | 101 |
| “Key”:       | 111 | 101 | 000 | 011 | 101 | 110 | 001 | 011 | 101 | 101 |
| “Plaintext”: | 001 | 000 | 100 | 010 | 011 | 000 | 110 | 010 | 011 | 000 |
|              | h   | e   | l   | i   | k   | e   | s   | i   | k   | e   |

- Try it out for
  - Key 3 = 1010111110101000111110000010001

# One Time Pad (OTP) (Cont...)

- OTP encryption scheme is provably secure
  - Ciphertext does not provide any information about the plaintext
  - All the derived plaintext are equally likely(heil hitler, kill hitler, heil kensike)
  - ~~Padding(assigning random letters to the numbers) is entirely up to the sender and the receiver~~
- Best Practice :
  - Pads should be used only once and randomly
  - Pads should be known only to the sender and the receiver

# One Time Pad (OTP) (Cont...)

- Drawbacks :
  - Pad / key length = plaintext length = ciphertext length
  - Key should be securely transmitted to the receiver before decrypting the message
  - If such secure transmission exist, what is the use of encryption
  - Message integrity ???
  - Depth issue
  - Guessing issue



# One Time Pad (OTP) (Cont...)

*Depth issue (Two ciphertext encrypted from same pad):*

e=000 h=001 i=010 k=011 l=100 r=101 s=110 t=111

- Plaintext 1 : like  $\rightarrow$  100 010 011 000
- Key : 110 011 101 111
- Ciphertext 1  $C_1 = P_1 \oplus K \rightarrow$  010 001 110 111  $\rightarrow$  ihst
- Plaintext 2 : kite  $\rightarrow$  011 010 111 000
- Key : 110 011 101 111
- Ciphertext 2  $C_2 = P_2 \oplus K \rightarrow$  101 001 010 111  $\rightarrow$  rhit

$$C_1 \oplus C_2 = P_1 \oplus K \oplus P_2 \oplus K = P_1 \oplus P_2$$

# Confusion and Diffusion

- By Claude Shannon – Founder of Information Theory (1949)
  - **Confusion** – Obscure relationship between plaintext and ciphertext
  - **Diffusion** – Spread plaintext **statistics** through the ciphertext
- 
- Confusion only                      ➔            simple substitution cipher, OTP
  - Diffusion only                     ➔            double transposition cipher
  - Modern day crypto systems are designed based on these concepts

# Cryptographic Methods

- Symmetric Key Crypto
- Asymmetric Key Crypto
- Hash Algorithms